

CloudVision AI — Research Report

Physics-Informed Hybrid Cloud Removal Pipeline Performance Analysis

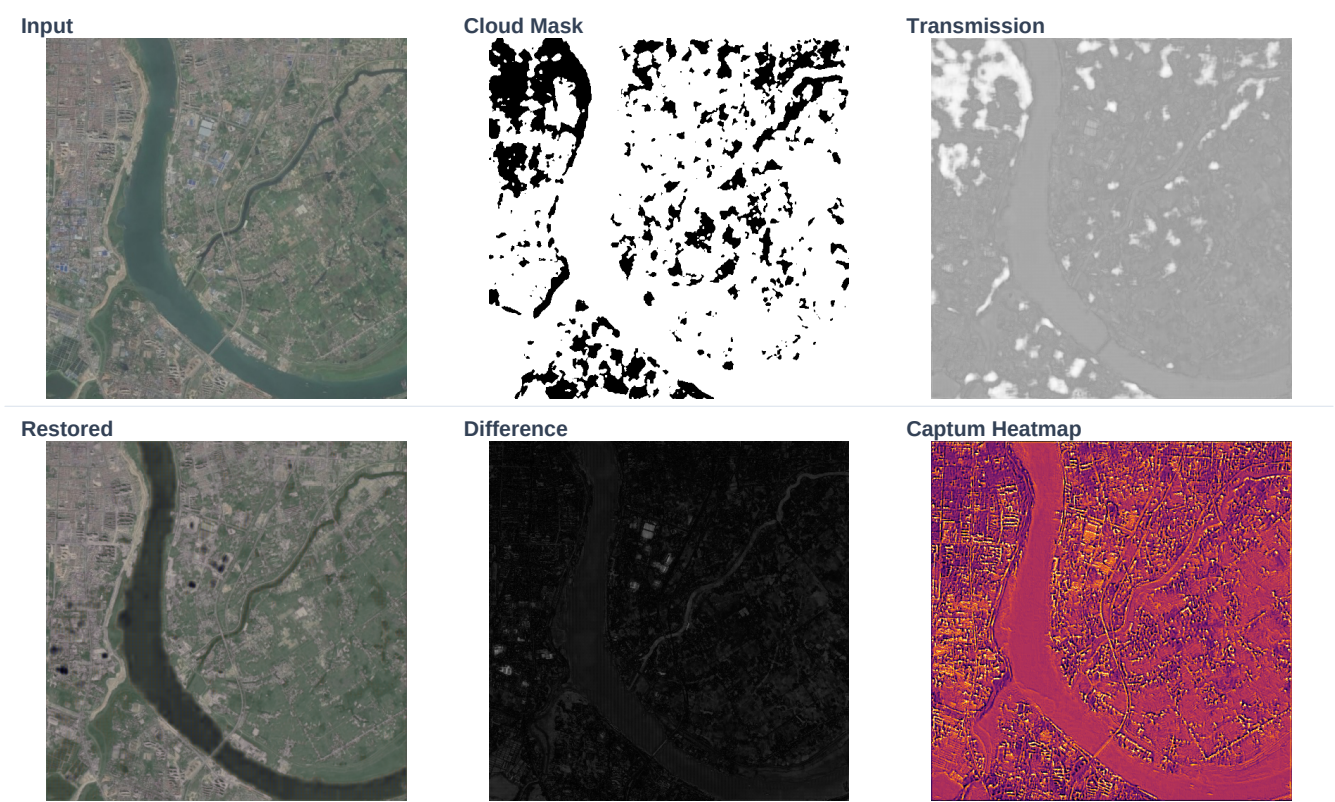
Executive Summary

This report documents the performance of the CloudVision AI Hybrid Cloud Removal Pipeline, which combines a U-Net semantic segmentation network with a Physics-Informed Neural Network (PINN) and Deep Image Prior (DIP) optimization. The model detects cloud cover, estimates atmospheric transmission and scattering parameters, and restores surface reflectance values while enforcing physical constraints (Atmospheric Scattering Model, Beer-Lambert attenuation, and NDVI consistency). Attribution heatmaps generated via Captum Integrated Gradients explain the feature importance.

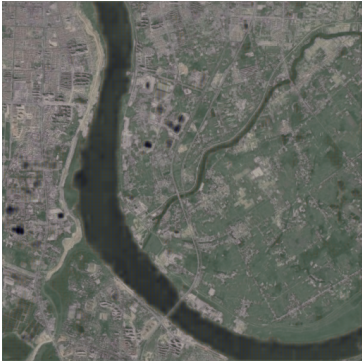
System & Processing Metrics

Pipeline Stage	Duration (sec)	Physics Loss	Value
U-Net Segmentation	0.32	Atmospheric Scattering (ASM)	0.0101
DIP Optimization	80.5	Radiative Transfer (RTE)	0.1022
PINN Physics Constraints	26.8	NDVI Consistency Loss	0.0000
Captum Attributions	6.1	Beer-Lambert Attenuation	0.0000
Total Pipeline Time	113.7	Total Physics Loss	0.0225

Visual Pipeline Flow Results



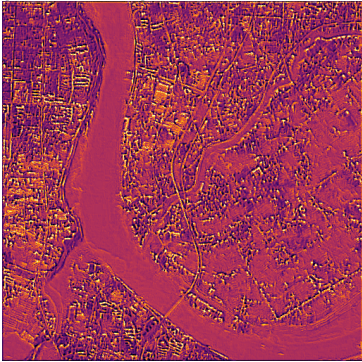
Restored



Difference



Captum Heatmap

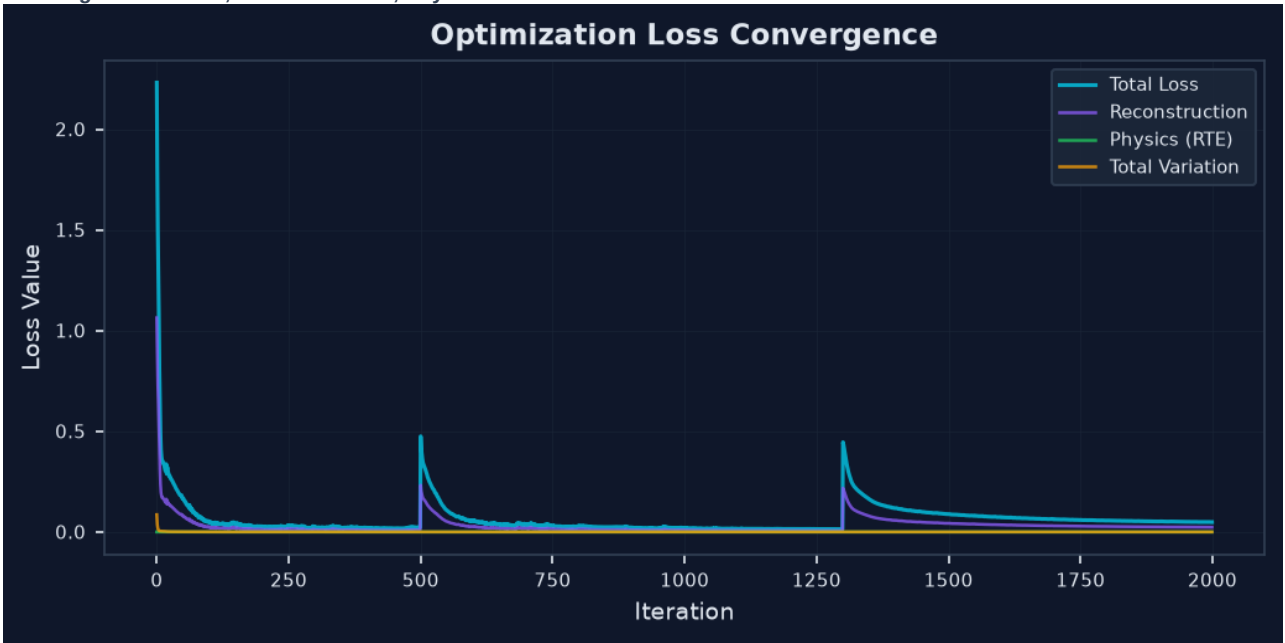


Validation & Quantitative Analysis

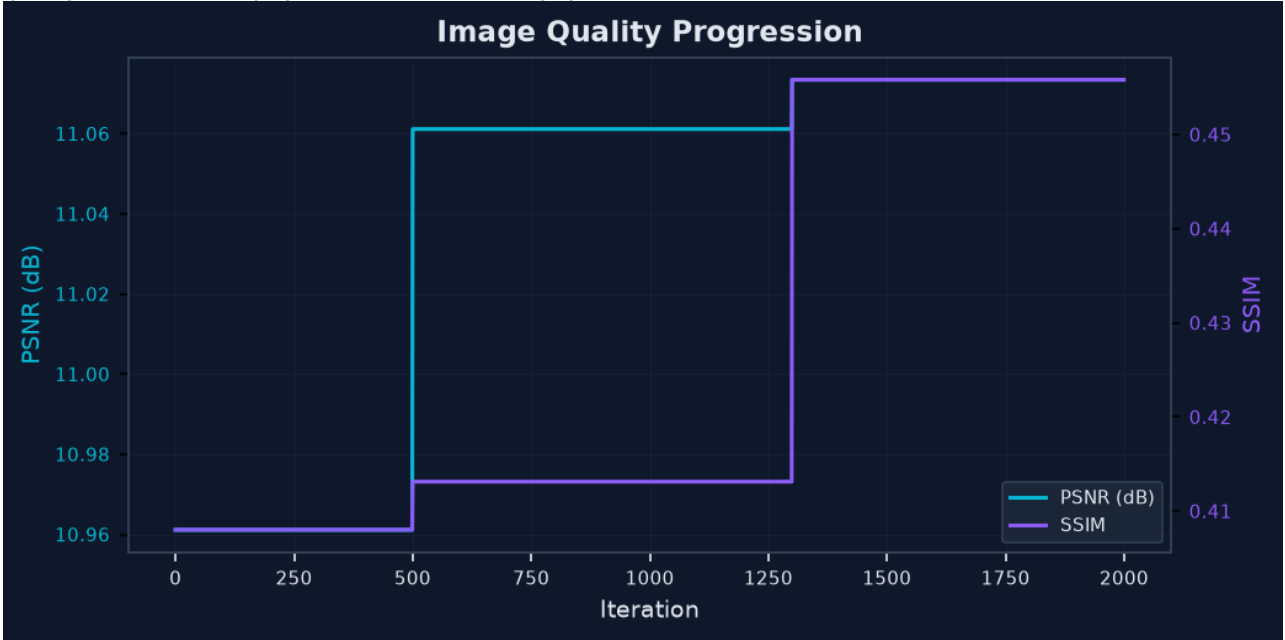
Segmentation Metric	Score	Restoration Metric	Value
Dice Similarity Coefficient	0.0000	PSNR (Peak Signal-to-Noise Ratio)	21.77 dB
Intersection over Union (IoU)	0.0000	SSIM (Structural Similarity)	0.6931
Segmentation Precision	0.0000	SAM (Spectral Angle Mapper)	0.1206
Segmentation Recall	0.0000	LPIPS (Perceptual Metric)	0.1520
F1 Score	0.0000	RMSE (Reflectance Error)	0.0816
Total Cloud Cover	80.1%	Cloud Detection Confidence	61.8%

Optimization Analysis Charts

Loss Convergence — Total, Reconstruction, Physics & TV losses over iterations



Quality Progression — PSNR (dB) and SSIM tracked during optimization



Cloud Type Distribution

Cloud Type	Coverage (%)
Thick Cloud	40.04%
Medium Cloud	24.02%
Thin / Cirrus	16.02%
Cloud Shadow	0.00%
Cloud-Free	19.92%

Scientific References

[1] Ulyanov, D., Vedaldi, A., & Lempitsky, V. (2018). Deep Image Prior. In CVPR.

[2] Raissi, M., Perdikaris, P., & Karniadakis, G. E. (2019). Physics-informed neural networks. Journal of Computational Physics.

[3] Kokhlikyan, N., et al. (2020). Captum: A unified model interpretability library for PyTorch. arXiv:2009.09996.

[4] Drusch, M., et al. (2012). Sentinel-2: ESA's multispectral imaging mission. Remote Sensing of Environment.